Voxel Structures Report

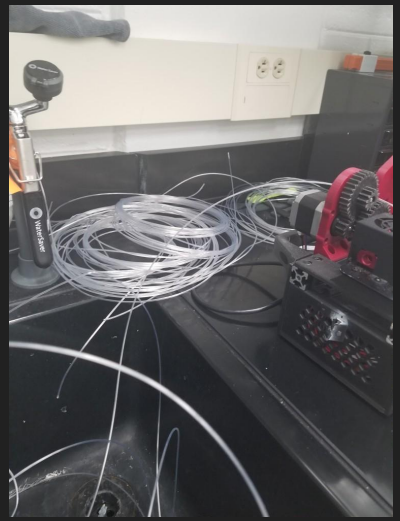
Shyam and Ephraim

Purpose of this Project

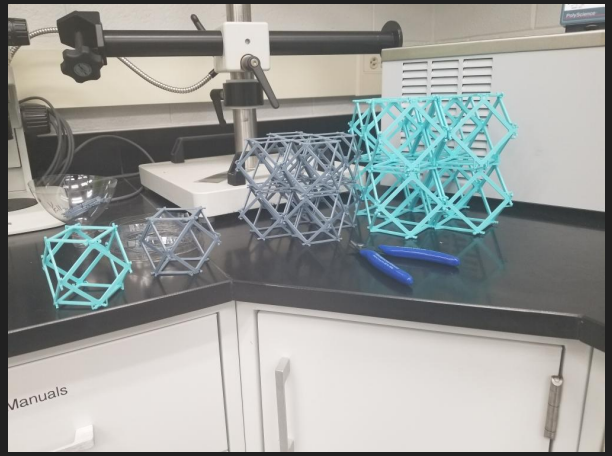
- Find a sustainable structural alternative to PLA/PETG
  - PET Development
    - Must have comparable material features to PLA/PETG
    - Must be printed and manufactured consistently
  - Voxel Design
    - Must be easily manufacturable
    - Must allow for sturdy connections



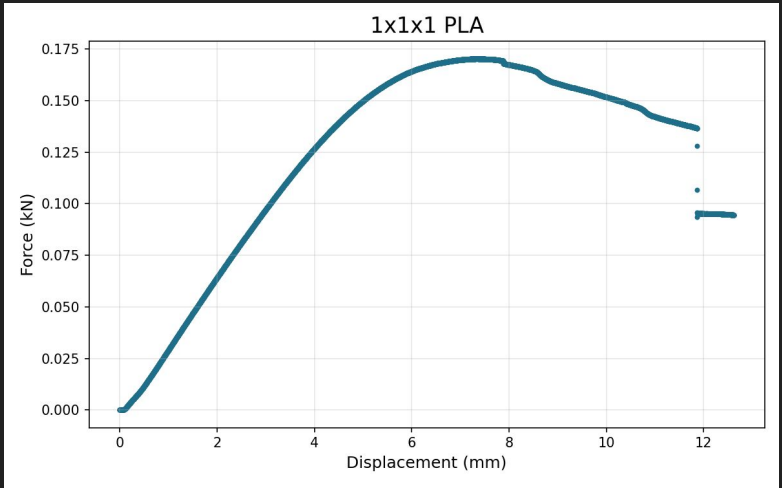
2L Plastic Bottle

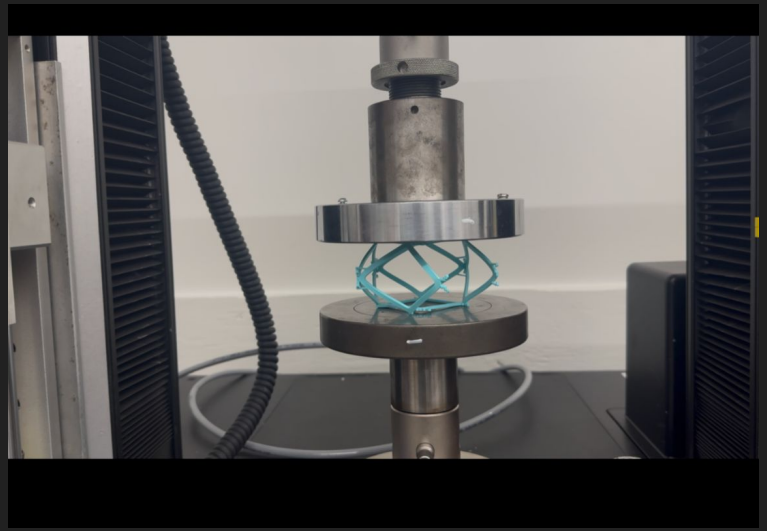


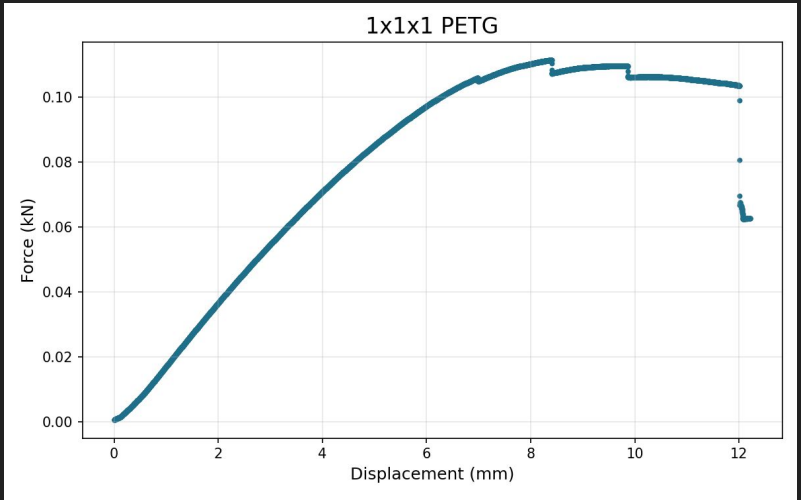
3D Printer Filament (PET)

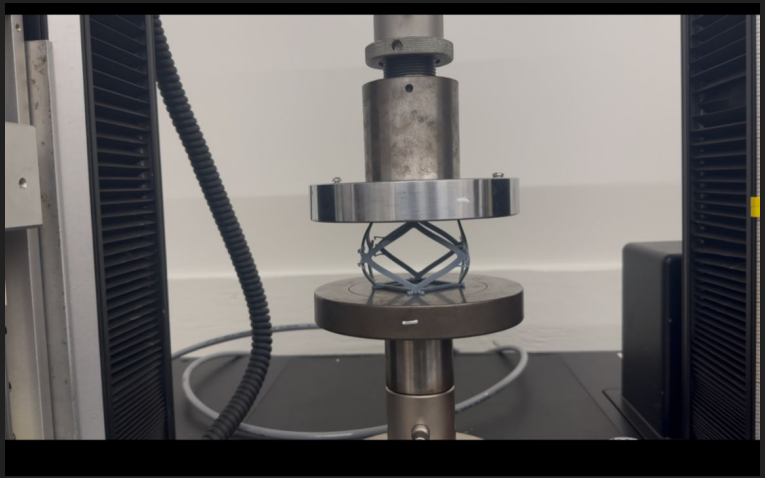


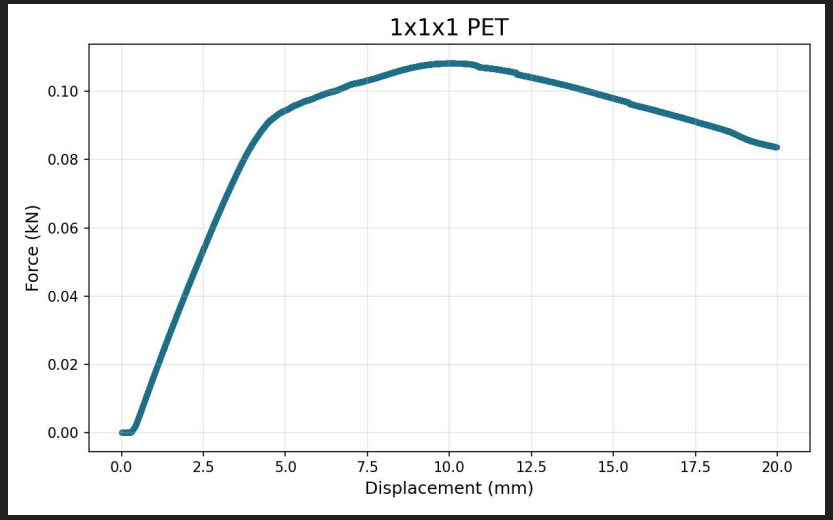
Voxel Structure

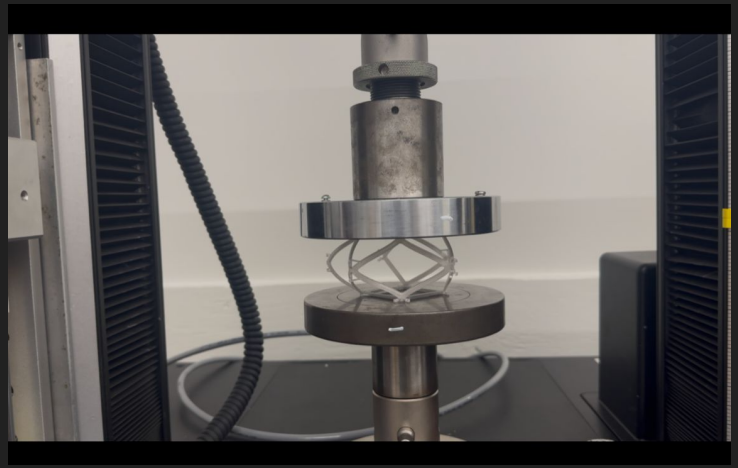

1x1x1 PLA
Force (kN)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
2
4
6
8
10
12
Displacement (mm)

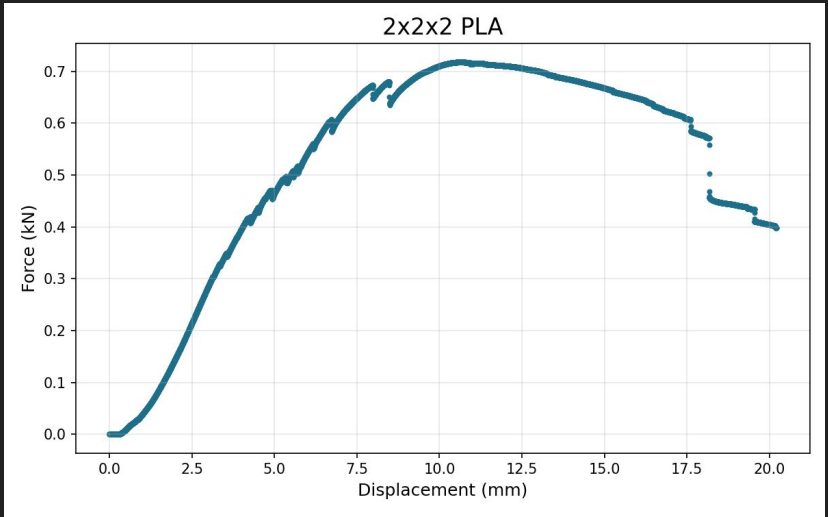


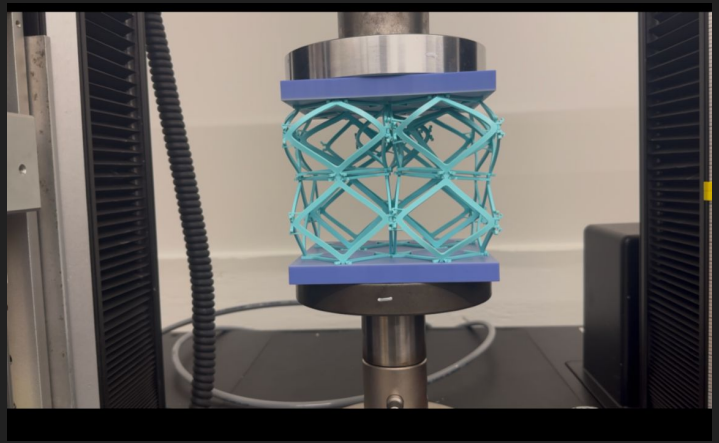

1x1x1 PETG
Force (kN)
0.00
0.02
0.04
0.06
0.08
0.10
0
2
4
6
8
10
12
Displacement (mm)

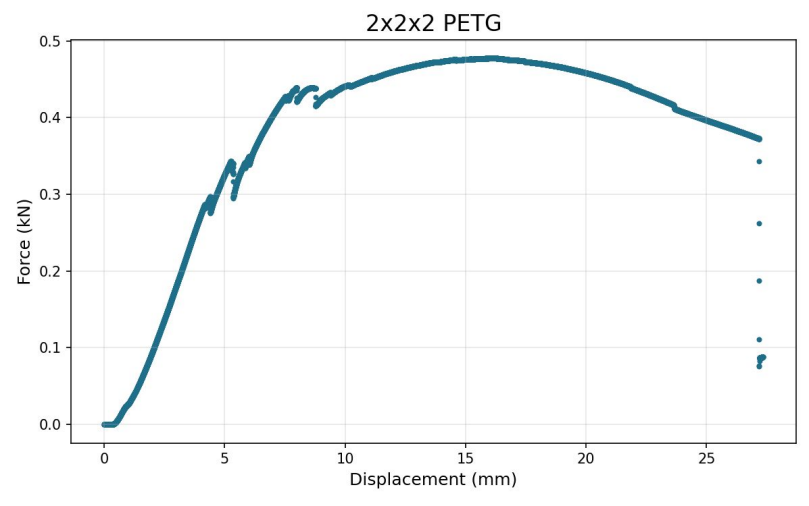


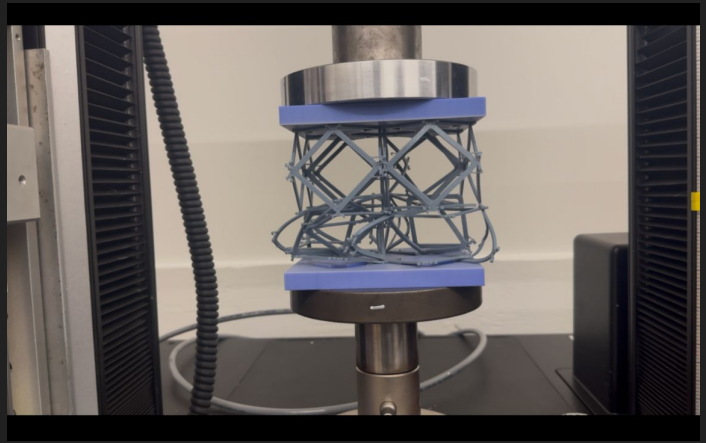

1x1x1 PET
Force (kN)
0.00
0.02
0.04
0.06
0.08
0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Displacement (mm)

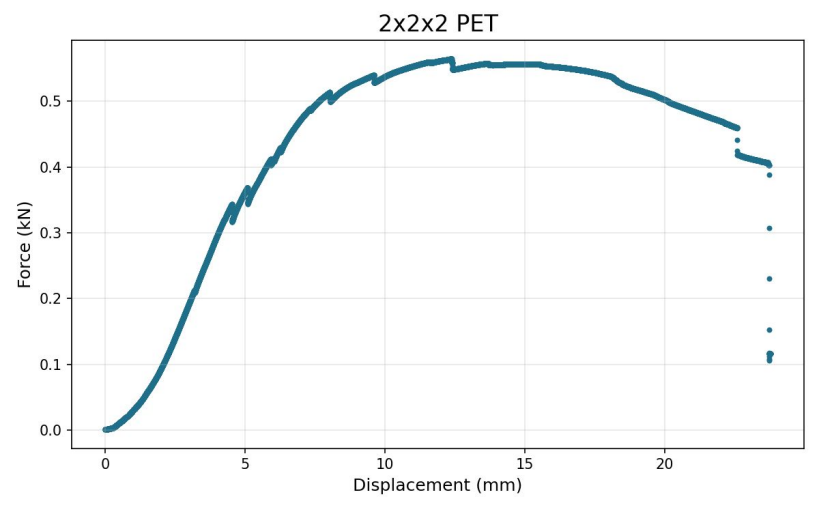


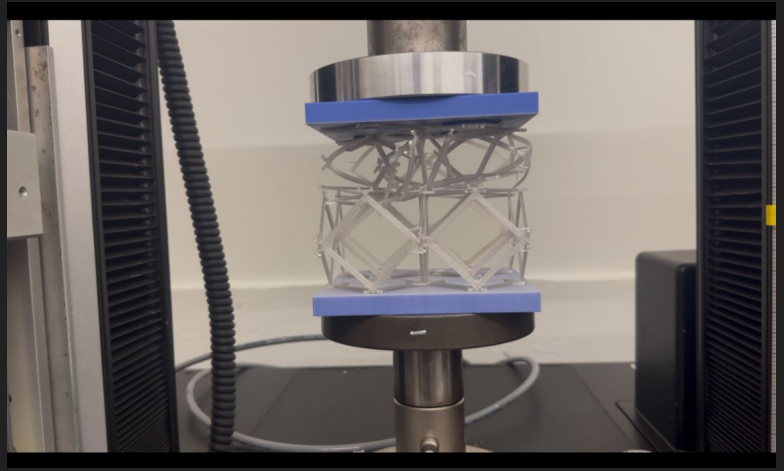

2x2x2 PLA
Force (kN)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Displacement (mm)

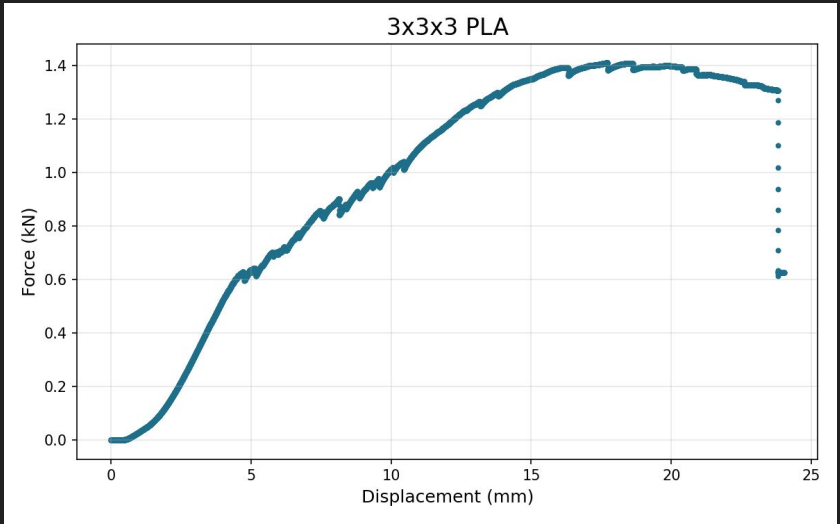


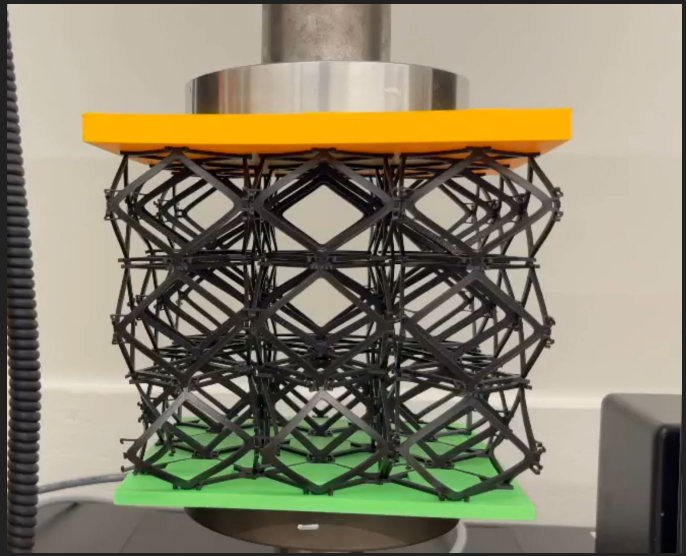

2x2x2 PETG
Force (kN)
Displacement (mm)
0.0
0.1
0.2
0.3
0.4
0.5
0
5
10
15
20
25

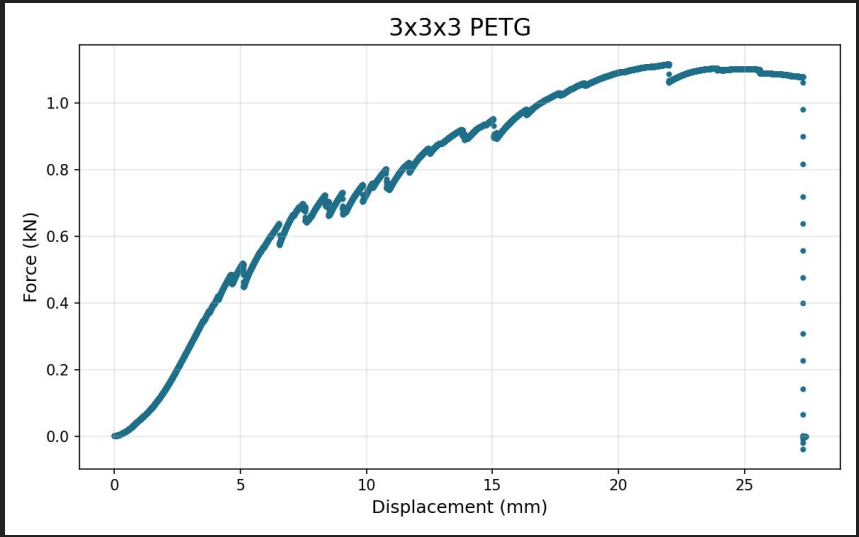


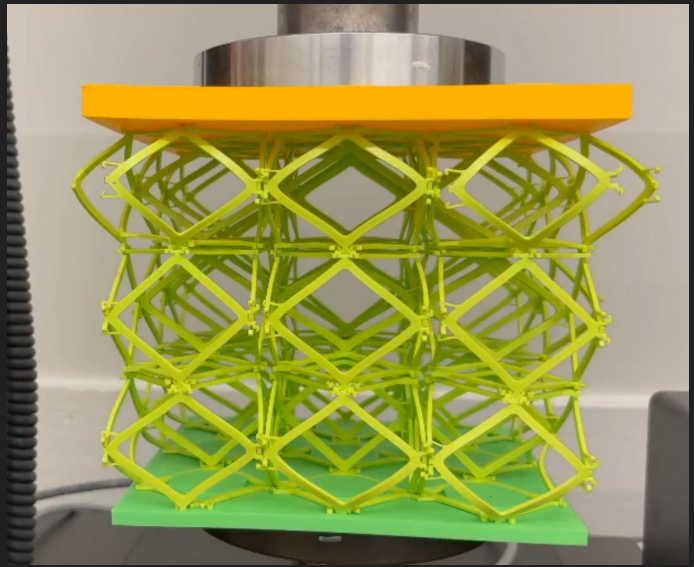

2x2x2 PET
Force (kN)
0.0
0.1
0.2
0.3
0.4
0.5
0
5
10
15
20
Displacement (mm)

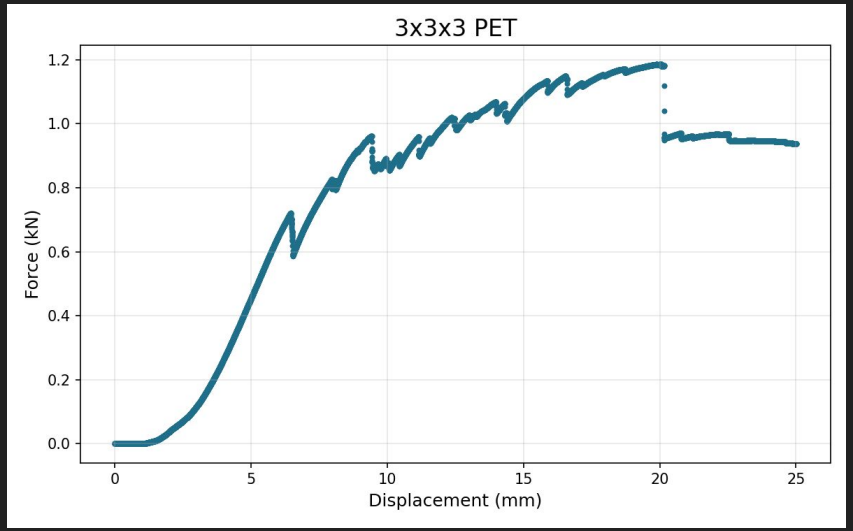


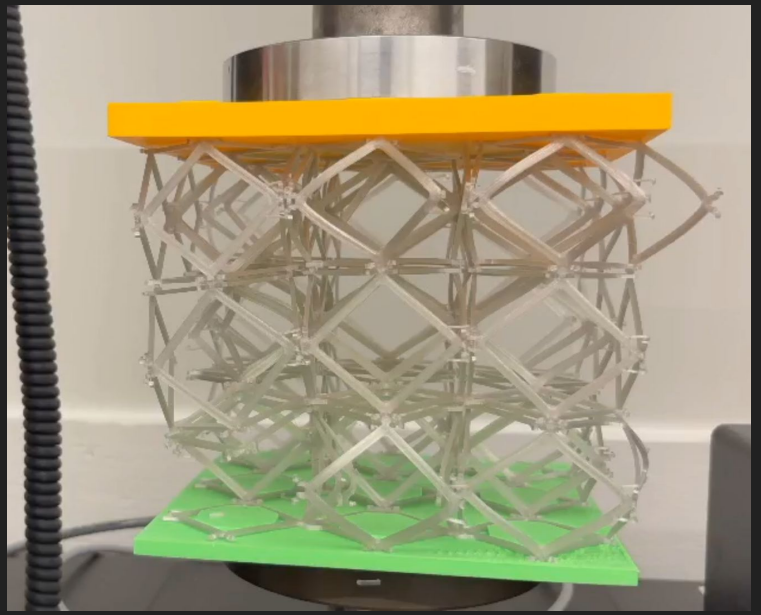

3x3x3 PLA
Force (kN)
Displacement (mm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
5
10
15
20
25

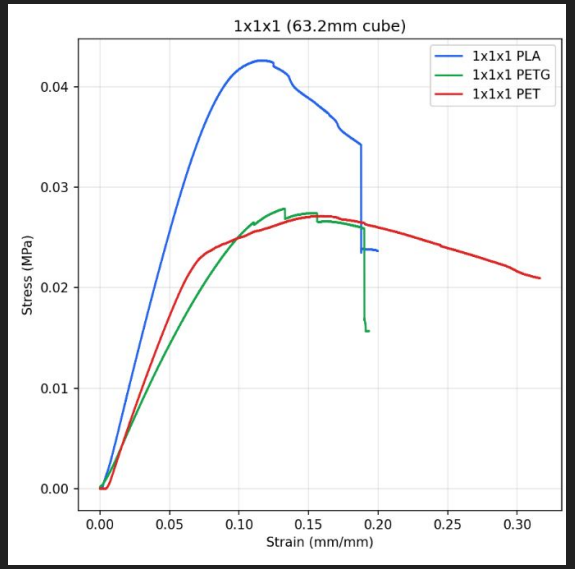


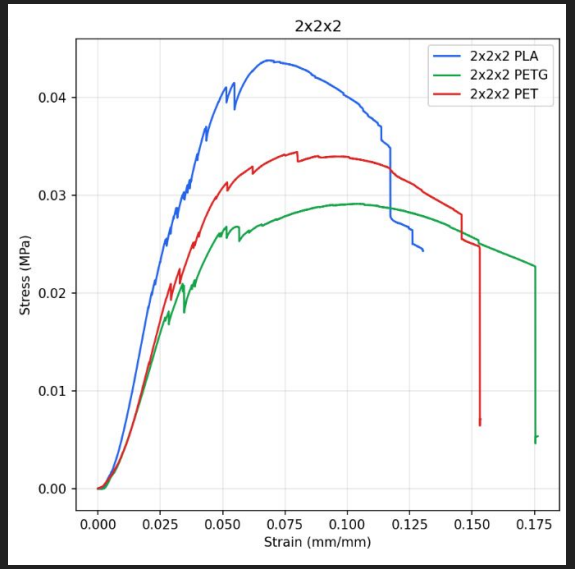

3x3x3 PETG
Force (kN)
Displacement (mm)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25

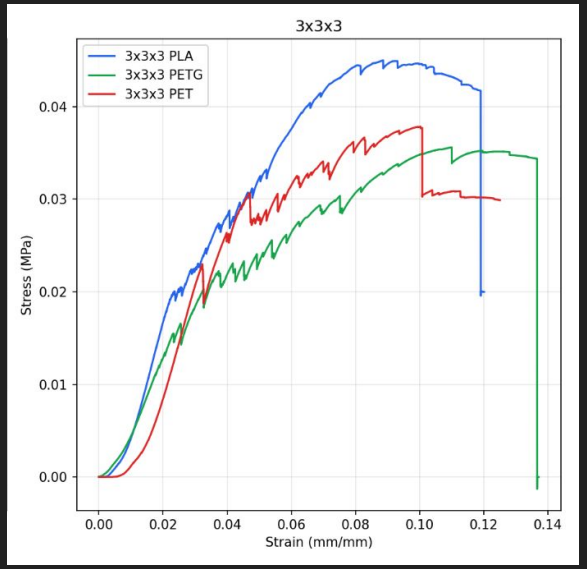



3x3x3 PET
Force (kN)
Displacement (mm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
5
10
15
20
25




1x1x1 (63.2mm cube)
1x1x1 PLA
1x1x1 PETG
1x1x1 PET
Stress (MPa)
Strain (mm/mm)
0.00
0.01
0.02
0.03
0.04
0.00
0.05
0.10
0.15
0.20
0.25
0.30

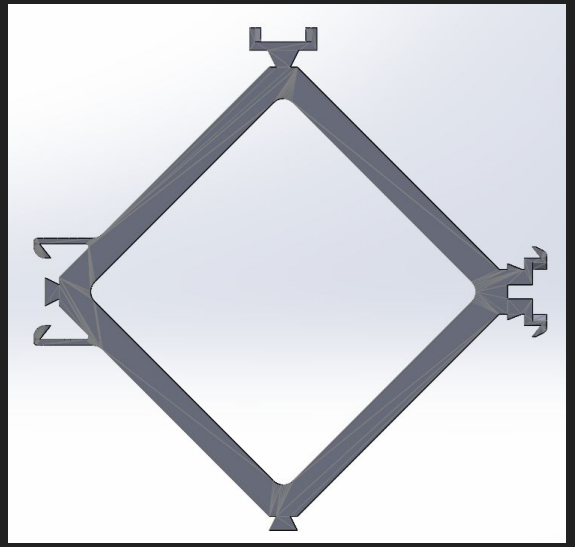

2x2x2
2x2x2 PLA
2x2x2 PETG
2x2x2 PET
Stress (MPa)
Strain (mm/mm)
0.00
0.01
0.02
0.03
0.04
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175

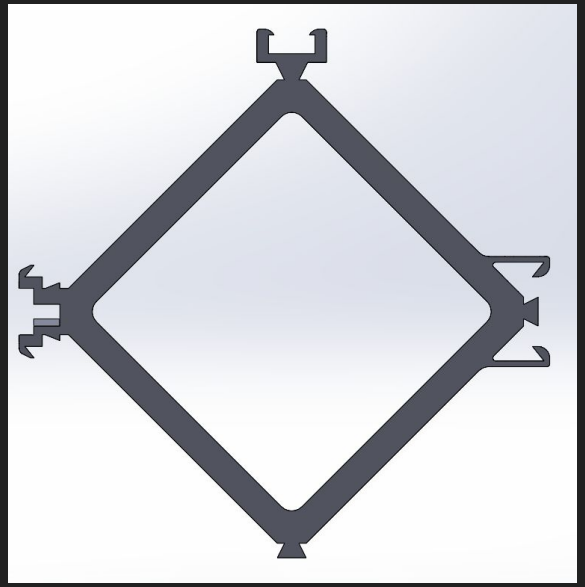

3x3x3
3x3x3 PLA
3x3x3 PETG
3x3x3 PET
Stress (MPa)
Strain (mm/mm)
0.00
0.01
0.02
0.03
0.04
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14

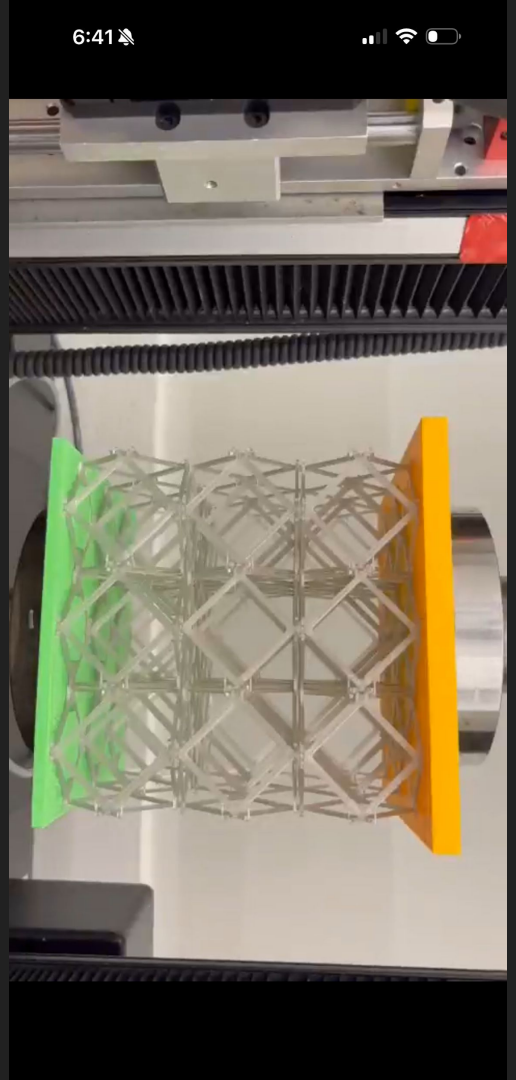
Future of Voxel Design

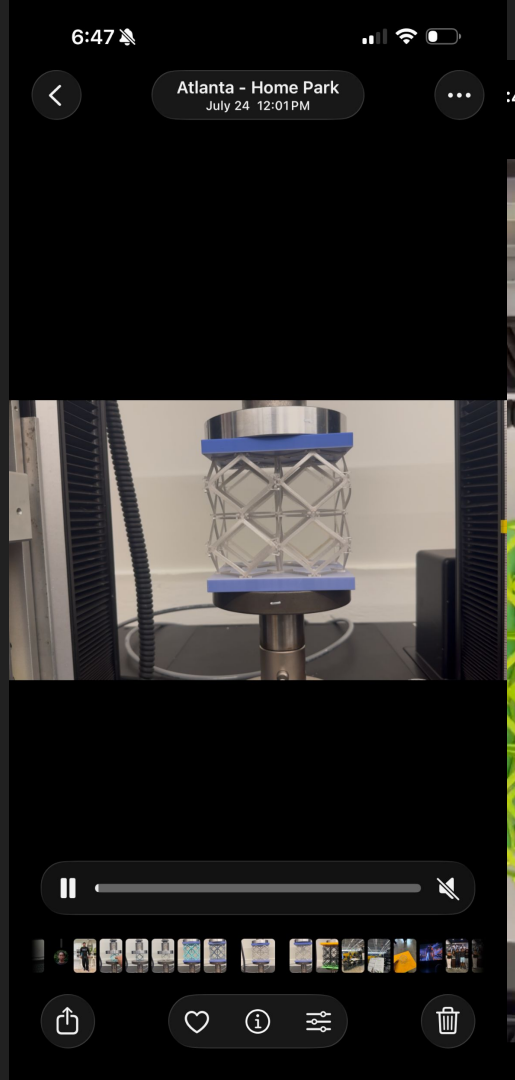
We've been researching to implement a new design for some of the voxel pieces

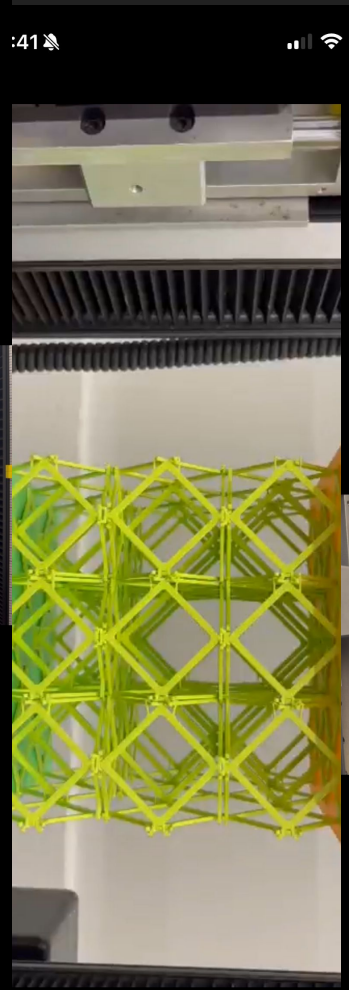
- Helps with connection between pieces
- Prints without many issues
- Doesn't change geometry/mass properties by a significant amount






6:41


6:47
Atlanta - Home Park
July 24 12:01 PM


:41


6:47
Atlanta - Home Park
July 24 11:49 AM